

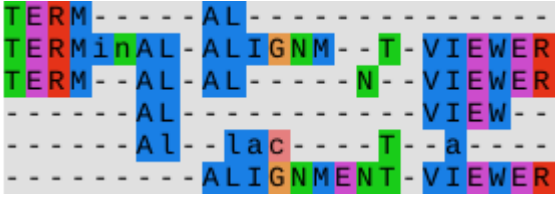
# Termal

## User Manual

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```
S. trutta schuberti
Elephas roseus
Buteo triplex
Oryctolagus paschalis
Anas laccata
Abies natalicia
```



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## Overview

**Termal** is a terminal-based viewer for multiple sequence alignments (MSAs). It is designed for fast, keyboard-driven navigation of large alignments, particularly in remote or SSH-based environments where graphical tools are impractical.

Termal is a **read-only** viewer. It does not modify alignments. This may change in future versions.

Termal is presented in (Junier 2025).

## Basic usage

### Starting Termal

Simply pass your alignment file as an argument to `termal`, e.g.:

```
termal my-alignment.msa
```

### Alignment File Formats

Termal supports Fasta (default) and Stockholm (pass `-f stockholm` (or just `-f s`)) formats. Any alignment lines shorter than the longest one will be padded with gap characters at the end.

## Help

For general help, pass option `-h`; for a list of key bindings, pass `-b` or press `?` after launching `termal`.

## Info mode

With option `-i/--info`, `termal` will output basic metrics about the alignment (such as number of sequences) and quit.

## Options

The main options are shown in the table below (and are discussed in the text in the corresponding sections). Other options exist, but they are either experimental or used for debugging or testing. For a full list, do `termal -h`.

| short           | long                                         | function                                                                                                       |
|-----------------|----------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| <code>-b</code> | <code>--show-bindings</code>                 | Show key bindings and exit successfully                                                                        |
| <code>-i</code> | <code>--info</code>                          | Show alignment metrics and quit                                                                                |
| <code>-f</code> | <code>--format &lt;FORMAT&gt;</code>         | Sequence file format [ <code>fasta/stockholm</code> ] (or just <code>f/s</code> ); default: <code>fasta</code> |
| <code>-o</code> | <code>--user-order &lt;USER_ORDER&gt;</code> | User-supplied order (filename)                                                                                 |
| <code>-c</code> | <code>--color-map &lt;COLOR_MAP&gt;</code>   | Gecos color map (filename)                                                                                     |
| <code>-n</code> | <code>--dry-run</code>                       | Dry run - show parameters and quit                                                                             |
| <code>-h</code> | <code>--help</code>                          | Print help                                                                                                     |
| <code>-V</code> | <code>--version</code>                       | Show version                                                                                                   |

## Screen layout

The interface is divided into four areas, starting from the top and left to right::

- **Headers pane** or simply **left pane**: shows sequence numbers and headers, as well as a barplot of the current [metric](#)
- **Alignment pane** or **main pane**: shows the aligned sequences, some properties of the alignments, as well as some current UI settings.
- **Corner pane**: shows the current metric and [ordering](#) (see below).
- **Reference pane** or **bottom pane**: shows horizontal position, the reference sequence (usually the consensus, but see [setting the reference](#)), and a conservation barplot

In addition, the last line contains a message area (“modeline”), which displays:

- pending command arguments (counts, searches), if any;
- search match information (when applicable).

## Interaction

`Termal` is entirely keyboard-driven. All commands are issued by typing a single character (usually a letter).

## Prefix arguments

Many commands (mostly motion) accept a *prefix argument*, which is an integer number typed *before* the command character. Typing any digit (except 0) starts a prefix argument. To cancel, press `Esc`. The meaning of the argument depends on the command. For example, scrolling commands such as `j` (“scroll one line down”) interpret a prefix argument as a repetition count: typing just `j` will move one sequence down, but typing `12j` will move twelve sequences down.

If no prefix argument is given, commands default to 1.

## String Arguments

String arguments are entered after typing the command character, and are entered by typing Return/Enter. Currently only the [header search](#) command (`"`), [sequence search](#) command (`/`) and [reference selection](#) command (`R`) take a string argument.

## Navigation fundamentals

Navigation in Ternal is inspired by [Vim](#) but simplified and adapted to multiple sequence alignments. The effect of motion commands depends on the [zoom mode](#), as follows:

- In zoomed-in mode, they change the portion of the alignment that is being displayed, much as a pager pages through a text file.
- In the zoomed-out modes, they move the zoom box.

Motion commands all take a prefix argument.

## Scrolling

Scrolling commands move the visible part of the alignment by one or more steps. The prefix argument is interpreted as a repeat count.

### Single-step scrolling

The following keys move the viewport by one step.

| command               | motion                                   |
|-----------------------|------------------------------------------|
| <code>[count]h</code> | scroll <i>count</i> columns to the left  |
| <code>[count]l</code> | scroll <i>count</i> columns to the right |
| <code>[count]j</code> | scroll <i>count</i> sequences down       |
| <code>[count]k</code> | scroll <i>count</i> sequences up         |

### Scrolling by screenfuls

To move by a screenful, use uppercase motion commands:

| command               | motion                                      |
|-----------------------|---------------------------------------------|
| <code>[count]H</code> | scroll <i>count</i> screenfuls to the left  |
| <code>[count]L</code> | scroll <i>count</i> screenfuls to the right |
| <code>[count]J</code> | scroll <i>count</i> screenfuls down         |
| <code>[count]K</code> | scroll <i>count</i> screenfuls up           |

## Aliases

The arrow keys are aliases for the motion keys.

| arrow | alias for      |
|-------|----------------|
| ←     | <code>h</code> |
| →     | <code>l</code> |

| arrow   | alias for |
|---------|-----------|
| ↓       | j         |
| ↑       | k         |
| Shift-← | H         |
| Shift-→ | L         |
| Shift-↓ | J         |
| Shift-↑ | K         |

The Space key is an alias for J (move one screen down). This mimics the behaviour of `less` and other Unix pagers.

### Examples

- h: move 1 column to the left
- 10l: move 10 columns to the right
- 5K: move 5 screens up
- 2<Spc>: move two screens down

## Jumps

Jump commands move directly to a position, specified by the prefix argument. The viewport moves so that the target sequence (respectively column) appears at the top (respectively left) of the alignment pane.

**NOTE:** in vertical jumps, the sequence's number is counted from the *first screen line*. This will be the same as the sequence's number in the alignment file, unless the sequences have been [reordered](#).

### Absolute jumps

In absolute jumps, the prefix argument denotes a specific sequence or column:

| command  | motion                                                 |
|----------|--------------------------------------------------------|
| [count]- | jump to sequence <i>count</i> (in the current order)   |
| [count]= | jump to sequence <i>count</i> (in original file order) |
| [count]  | jump to column <i>count</i>                            |

### Relative jumps

In relative jumps, the prefix argument denotes a percentage of the alignment's height or width, rounded to the nearest sequence/column.

| command  | motion                                           |
|----------|--------------------------------------------------|
| [count]% | jump to <i>count</i> % of the alignment's height |
| [count]# | jump to <i>column</i> % of the alignment's width |

### Jumps to search matches

Termal supports regular expression searches in both headers and sequences (see [Searching](#) for how to start a search). If any matches are found, Termal will automatically jump to the first match. To jump to the next or previous match, use n or p:

| command  | motion                                             |
|----------|----------------------------------------------------|
| [count]n | jump <i>count</i> matches forward                  |
| [count]p | jump <i>count</i> matches backward                 |
| <Return> | jump to the current match (which may be offscreen) |

Next and previous match jumps wrap around, i.e. pressing n while on the last match will move back to the first one. If no matches were found, match jump commands have no effect.

## Examples

- 123|: Jump to column 123 (if it exists).
- 200-: Jump to the 200th sequence from the top of the display, which may differ from the original file order if sequences have been reordered.
- 50%: Jump to the vertical midpoint of the alignment.
- 25#: Jump to one quarter of the alignment width.
- n: jump to the next match
- 3p: jump three matches backward

## Zooming

In Ternal, *zooming* refers to changing how the alignment is displayed, rather than scaling characters graphically (although this can be done by changing the terminal emulator's font size). Zooming is only relevant if the whole alignment does not fit on screen. It is possible for the alignment to fit on the screen only in one of the two dimensions: in this case, the zooming only applies to the other dimension.

In *zoomed-in* mode, Ternal displays as many *adjacent* sequences and columns as will fit on the screen. This shows a detailed view of a portion of the alignment, but obscures its large-scale features. Motion commands change which portion of the alignment is displayed, causing new sequences/columns to appear into view and as many sequences/columns to disappear. Ternal starts in zoomed-in mode.

In *zoomed-out* mode, Ternal shows the first and last sequence/column as well as a uniform sampling of sequences/columns between them. Enough sequences/columns are sampled to fill the screen; the other sequences/columns are not shown. This shows a “big picture” view of the alignment, at the expense of granularity. Motion commands affect the part of the alignment that will be shown when moving back to zoomed-in mode. This area is shown on screen as a rectangle known as the *zoom box*.

A variant of the *zoomed-out* mode works as above but preserves the alignment's aspect ratio. It is called *zoomed-out-AR*.

To cycle forward through the zoom modes, press z; to cycle backward, press Z.

## Searching

### Searching sequence headers

Ternal supports searching within sequence headers using regular expressions.

To start a search:

1. Press " (double quote)
2. Enter a [regular expression](#)
3. Press <Enter> to confirm

To cancel a search, press <Esc>.

During a search, the modeline displays:

- the index of the current match,
- the total number of matches.

**Example:**

```
"^Eco<Enter>
```

This jumps to the first sequence whose header starts with Eco (if any). The search order is according to the current [ordering](#).

## Searching sequences

Termal supports searching within sequences using regular expressions.

To start a search:

1. Press /
2. Enter a [regular expression](#)
3. Press <Enter> to confirm

To cancel a search, press <Esc>.

During a search, the modeline displays:

- the index of the current match,
- the total number of matches.

**NOTE** The current implementation of sequence search does not treat gap characters specially. So if you want e.g. to specify a possible gap between A and T, you need to do so explicitly: A-?T. A smarter handling of gaps is planned for a future release.

**Example:**

```
/GAATTC<Enter>
```

This jumps to the first instance of GAATTC. The search order is according to the current [ordering](#), then left to right.

## Setting the Reference

Some features of the alignment are computed with respect to a *reference sequence*. This is the case, for example, of the similarity metric, which measures how similar each sequence is to the reference.

By default, the reference is the consensus sequence (which is automatically computed). However, the user may select any sequence in the alignment to serve as reference. This is done with the R command, which takes a sequence number (with respect to the original file) as an argument. If the sequences are currently ordered according to the similarity metric, changing the reference triggers a recomputation of the ordering.

Passing an empty argument reverts to the consensus.

**Example:**

```
R12<Enter>
```

This selects sequence #12 as reference. This is reflected in the corner panel, which now reads 'Ref: #12', and in the bottom pane, which displays sequence #12 instead of the consensus.

## Display Controls

### Resizing the Left Pane

The left pane can be widened (perhaps to show more of the sequence headers) with `>` and shrunk with `<`. This also resizes the corner pane. Both accept a prefix argument, which is by how many characters the pane is to be resized:

| command                  | motion                                          |
|--------------------------|-------------------------------------------------|
| <code>[count]&gt;</code> | widen the left pane by <i>count</i> characters  |
| <code>[count]&lt;</code> | shrink the left pane by <i>count</i> characters |

### Ordering the Sequences

By default, the sequences appear in the alignment pane in the same order as they appear in the alignment file. However, the sequences may be ordered according to the current [metric](#), either ascending or descending. Note that the “first” sequence (on the screen) is the *top* sequence. As a result, when sorting in ascending order, metric values increase from top to bottom.

The user may also supply a custom ordering, by passing option `-o` and supplying the name of an ordering file (see below) as an argument to the option, e.g.:

```
termal -o my-order alignment.msa
```

In this case, `termal` initially shows the alignment in custom ordering.

### Ordering File

An ordering file is simply the sequence headers in the desired order. For example, passing the following ordering file would cause `AHMKMHDK_00298` to appear first, then `CEFNMEKK_03699`, etc., regardless of the order they appear in in the alignment file.

Note that the headers in the ordering file are expected to match those in the alignment file.

```
AHMKMHDK_00298
CEFNMEKK_03699
IAGGDKJC_03995
FHL0DNDD_02091
HJACH0PP_04370
```

### Changing the Ordering

To cycle forward through orderings, press `(o)`. To cycle backward, press `0`.

The current ordering is displayed in the corner pane, as a symbol just below the bar chart in the left pane:

| symbol         | meaning                                 |
|----------------|-----------------------------------------|
| <code>-</code> | file order                              |
| <code>↑</code> | current metric, ascending               |
| <code>↓</code> | current metric, descending              |
| <code>u</code> | user-supplied order ( <code>-o</code> ) |



## Metrics

The left pane displays a bar chart of the current *metric*. This is a numeric property of the sequences. Currently, there are two possible metrics:

- Sequence length (not counting gaps)
- Similarity to the [reference](#)

To cycle forward through the metrics, press `t` (metric); press `T` to cycle backward. The current metric is displayed in the corner pane.

The sequences can be [ordered](#) according to the current metric.

## Residue Colormaps

Termal supports four built-in residue color maps:

| source     | residue class | reference                |
|------------|---------------|--------------------------|
| ClustalX   | amino acids   | (Larkin et al. 2007)     |
| Lesk       | amino acids   | (Lesk 2019)              |
| JalView    | nucleotides   | (Waterhouse et al. 2009) |
| monochrome | both          | <a href="#">themes</a>   |

It is also possible to supply a custom color map:

```
termal -c colormap.json my-alignment.msa
```

where the colormap is a JSON file in the Gecos format (Kunzmann et al. (2020)). This is a straightforward format that looks like this:

```
{
  "name": "my-colormap",
  "alphabet": [
    "A",
    "C",
    "G",
    "T",
  ],
  "colors": {
    "A": "#71564e",
    "C": "#2a3d00",
    "G": "#004f7c",
    "T": "#b03f42",
  }
}
```

To cycle through colormaps, type `m` (forward) or `M` (backward). The initial colormap is ClustalX for amino acids or JalView for nucleotides, unless a custom colormap is supplied, in which case it becomes the initial colormap. The current colormap is displayed in the top border of the Termal screen.

## Themes

Termal supports three themes: dark, light, and monochrome. To cycle through the themes, type `s` (forward) or `S` (backward). The current theme is displayed in the top border of the Termal screen.

**NOTE** Termal has been tested predominantly in a dark-themed terminal.

## Inverse Video

By default, Termal displays the sequence residues in inverse video. To toggle to direct video (and back), press **i**. The current video mode is displayed in the top border of the Termal screen.

## Modeline and feedback

The modeline (in the bottom border of the Termal screen) provides feedback about: - pending numeric arguments, - active search state, - current search match index.

---

## Limitations and scope

Termal currently does **not** support:

- editing alignments
- graphical export
- alignment formats other than Multi-fasta and Stockholm

## Future Work

Features planned for the next release:

- better handling of gaps in regexp searches
- jumps to conserved regions
- a color map for vision-impaired users

Features that will be added later

- exporting (part of) the alignment as SVG
- arbitrary sequence as reference
- showing only residues that differ from the consensus, or that are fully conserved (à la EMBOSS's `showalign -show`)

Features under consideration

- Showing a phylogeny in the left pane
  - Simple edits (removing empty columns)
  - Saving parts of the alignments
- 

## Design philosophy

Termal prioritizes:

- predictability over feature breadth,
- keyboard-driven navigation over menus,
- robustness over visual effects.

It is intended for users comfortable with terminal-based tools who value speed and clarity over graphical interaction.

Many commands, as well as the prefix argument syntax, were deliberately copied from [Vim](#).

---

## Companion programs

### termal-export

This program produces SVG representations of alignments, e.g. for publication. Its call syntax is simply:

```
termal-export [options] <alignment file>
```

The SVG is printed out to standard output, so it's usually best to redirect it to a file of your choice, e.g.

```
termal-export my-aln.msa > my-aln.svg
```

### Options

The main options supported by termal-export are shown in the table below. Other options exist, but they are either experimental or used for debugging or testing. For a full list, pass -h.

| short | long                            | function                                                                                         |
|-------|---------------------------------|--------------------------------------------------------------------------------------------------|
| -o    | --order <ORDER>                 | User-supplied order file (as in termal) (TODO)                                                   |
| -c    | --colormap-name <COLORMAP_NAME> | Colormap [default: aa-clustalx] [possible values: aa-lesk, aa-clustalx, dna-jalview, monochrome] |
|       | --rows <ROWS>                   | Row range as START:END (0-based, END exclusive). Example: 0:50 (TODO)                            |
|       | --cols <COLS>                   | Column range as START:END (0-based, END exclusive). Example: 0:200 (TODO)                        |
| -h    | --help                          | Print help                                                                                       |

## References

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